

HAYEON LEE, Ph.D.

Research Scientist — Retrieval, Post-training & Search Agents

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EXPERIENCE

Research Scientist | Meta — Meta Superintelligence Labs (MSL), Search Team | Oct 2024 – Present

Text Dense Retrieval

- Owned the full model lifecycle for text retrieval—data, training, evaluation, and production—and delivered the team's largest H1 2026 retrieval-quality gains.
- Rebuilt training data via strategic sampling (62% duplicates, 98% keyword-style, Hindi under- → 50% over-representation, decontamination): recall@5 +4.22pp/+13.9% (keyword), +7.60pp/+67.9% (conversational); rescued 25.1% zero-recall queries.
- Applied positive masking to remove false-negative collisions (zero data loss): recall eval @5 +2.31%.
- Increased effective batch size 384 → 3,094 for multi-GPU contrastive learning; offline Recall@50 +3.24 (+11%), Recall@300 +2.59 (+6.5%).
- Fixed entity-mismatch failures in the launched text model: entity recall@50 +1.66× via LLM-generated hard negatives; main POC for prompt optimization, training/eval datasets, and 13 model variations.
- Cut Google label dependency: LLM-generated training labels + Google-independent pipeline, launched as next model.

Image Dense Retrieval

- Built image dense retrieval 0 to 1: selected the backbone via zero-shot comparison of open-source retrievers and built a 1M/50M quick recall eval pipeline for fast iteration.
- Mixed prod traffic with open-source data to beat zero-shot (+2.5pp end-to-end, +26.1% recall relative), fixing fine-tuning regression via early stopping; contributed to the image search prod launch (Muse's search API), where Meta AI beat Bing image search on internal end-to-end eval.
- Launched 4 Image Search Early Stage Ranker (ESR) wins: embedding replacement (+4.9pp golden), 28-feature combo (+4.3pp freshness), dense/sparse rebalance (+3.52pp golden), AI-generated-content detection (0.955 AUROC).
- Launched Text-Text dense retrieval lane in Image Search production (~7.97B-doc context index); rebalanced Dense/Sparse/Context to 63/13/20 within budget.
- Shipped richer-context V2 index (22.60B docs exact-rendered, 17.64B embeddings verified, 7.80B matching V1): +1.23pp pass rate (5 evals), IQ +0.022, 0 errors at 1,098 QPS (p95 84.5ms).

Agent & RL Research

- Proposed Cycle-Consistent Search for gold-free search-agent training (question reconstructability as proxy reward; info bottlenecks + GRPO): beat strongest gold-free baseline by 4.5–9.8% and gold-supervised Search-R1 by +0.5/+1.3 pts (Qwen2.5-7B/32B). Accepted to CoLM 2026.
- Co-developed PastForward with UNIST: reused on-device VLM computation (verified multi-token proposals, GUI-transition prediction, KV-state carryover) with no extra fine-tuning; cut action-step latency 1.63–2.36× (edge) and 1.40–1.68× (server) at equal task success (20 apps, 116 tasks, 3 VLMs).

LLM-as-a-Judge Evaluation

- Defined URL- and response-level locality as core quality metrics; replaced \$80K-per-round human eval with an LLM-as-a-judge (hundreds/day) at 98/98 precision/recall vs. humans, mainly via prompt optimization.

Ranking

- Built 0-to-1 vertical ranker and launched Locality Reranker (gap vs. Google/Bing ~0.8–1.0% → 0.1–0.4%); co-led QE4/QEv1 launches (+0.97% helpfulness, +1.38% recall@5, +7.3% authority; PyTaco +8% Direct Authority).
- Distilled 405B → 70B LLM for Need Met label generation at equal quality (helpfulness 88.45 vs. 88.47) and 6× throughput (109.56 → 658.18 labels/min), unlocking billion-scale ranking training labels.
- Pruned legacy training labels via ablation (+4.81% offline authority, –6.47% freshness failures); retuned label weights 25 → 12 (+5.5% domain authority).
- Fine-tuned the ranker on domain-authority/freshness objectives: negative user feedback –2.7% (±1.6), positive feedback +0.92% (±0.60), +6.3% overall.

Postdoctoral Researcher | Meta FAIR Labs | Sep 2023 – Sep 2024

Advisor: [Yuandong Tian](#)

- Developed a retrieval-head approach to compress key-value caches for memory-efficient long-context factual generation.

SELECTED PUBLICATIONS

- [PastForward: Faster On-Device GUI Agents via Computational Experience Reuse](#). Arxiv 2027.
- [Cycle-Consistent Search: Question Reconstructability as a Proxy Reward for Search Agent Training](#). CoLM 2026 (accepted).
- [FRESCO: Benchmarking and Optimizing Re-rankers for Evolving Semantic Conflict in Retrieval-Augmented Generation](#). Findings of EMNLP 2026 (accepted).
- 8 additional first-author papers at top-tier venues: ICLR ×4 (2020 oral), NeurIPS ×2 (2021 spotlights), ACL/EMNLP Findings.

EDUCATION

Ph.D. | KAIST | 2018–2023 | Advisor: [Sung Ju Hwang](#) | Best Ph.D. Dissertation Award (2024)

M.S., KAIST (2016–2018); B.S., Computer Science, Sungkyunkwan University (2012–2016)